

torch.narrow_copy#

https://pytorch.org/docs/stable/generated/torch.narrow_copy.html

Description:

Same as `Tensor.narrow()` except this returns a copy rather than shared storage. This is primarily for sparse tensors, which do not have a shared-storage narrow method. `input` (Tensor) – the tensor to narrow `dim` (int) – the dimension along which to narrow `start` (int) – index of the element to start the narrowed dimension from. Can be negative, which means indexing from the end of `dim` `length` (int) – length of the narrowed dimension, must be weakly positive

Function Signature

```
torch.narrow_copy(input, dim, start, length, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> x = torch.tensor([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
>>> torch.narrow_copy(x, 0, 0, 2)
tensor([[ 1,  2,  3],
        [ 4,  5,  6]])
>>> torch.narrow_copy(x, 1, 1, 2)
tensor([[ 2,  3],
        [ 5,  6],
        [ 8,  9]])
>>> s = torch.arange(16).reshape(2, 2, 2, 2).to_sparse(2)
>>> torch.narrow_copy(s, 0, 0, 1)
tensor(indices=tensor([[0, 0],
                       [0, 1]]),
      values=tensor([[[[0, 1],
                       [2, 3]],
                      [[4, 5],
                       [6, 7]]]]),
      size=(1, 2, 2, 2), nnz=2, layout=torch.sparse_coo)
```

Notes & Warnings

NOTE

See also `torch.narrow()` for a non copy variant

Detailed Documentation

Documentation

Same as `Tensor.narrow()` except this returns a copy rather than shared storage. This is primarily for sparse tensors, which do not have a shared-storage narrow method.

`input` (Tensor) – the tensor to narrow

`dim` (int) – the dimension along which to narrow

`start` (int) – index of the element to start the narrowed dimension from. Can be negative, which means indexing from the end of dim

`length` (int) – length of the narrowed dimension, must be weakly positive

`out` (Tensor, optional) – the output tensor.

`torch.narrow()` for a non copy variant